Presentation 2

- Links to presentation(s) and code(s) on GitHub

Presentation:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Call%20outcome%20analysis/CallCategorization_Oct23_Group2.pdf

Code:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/IdentifyingAbandonedCalls_23Oct_Vivienne.ipynb

- What did you do?

For this presentation, my objective was to identify scenarios where call outcomes can be categorized as positive, negative, or neutral so that eventually the activity as the customer leaves can be visualized. To begin this, I looked at possible values for termination reasons included in the dataset and tried to identify which could be easily mapped to the three outcome categories. Then, I tried to identify call outcomes that either would not be reflected in the termination reason (e.g. for a courtesy callback, it is likely that the customer left their original call after selecting courtesy callback, but there then could be several attempts to reach them before they successfully speak to an agent so there would be several termination reasons for one 'Contact Session ID' and categorizing the call outcome would need to incorporate additional factors). Based on these scenarios, I then began to try and identify what additional factors would be needed to categorize the outcomes that did not match exactly with a termination reason. In particular, 'Customer Left' appeared to be the termination reason that had the most scenarios that needed further exploration. To begin this, I chose to focus on abandoned calls which I considered to be negative outcomes where the termination reason was 'Customer Left,' and I developed additional scenarios specifically for abandoned calls, building on the cases that had been identified in our mentor meeting and the examples proposed by Meera and Sana (Simple EDA Team 1). After identifying the different cases, I began trying to find a method identifying these scenarios in the dataset that can hopefully later be applied to visualizing the different outcomes. Since it appeared that the duration of each step in the journey and the overall call would be helpful in identifying if the call was abandoned, using the CAR dataset filtered to only include 'Contact Session IDs' where the termination reason was 'Customer Left,' I added a duration column and also incorporated Meera's data processing that added a 'Time Difference' column for each step of a caller's journey and the 'Call ID' column.

The first case I considered was the call being abandoned while the hold music was playing. I began by filtering my dataset to only include 'Contact Session IDs' that had 'PlayMOH300s' as the 'Activity Name' in at least one of its rows. I then sorted the dataframe by 'Call ID' and 'Activity Start Timestamp' so all rows for each call would be successive in the dataframe according to the time of each step. An example call from this dataframe is included in my code for this presentation. To find cases where the caller left during the hold music, my methodology was to identify cases where listening to hold music was in the row immediate prior to the call termination (i.e. it would have to be the caller's last activity so their issue could not have been resolved) and the 'Time Difference' between the last two steps was less than 300 seconds so they did not listen to the music in its entirety. I first wrote an 'if' statement based on these two conditions and utilized the index of an example I had identified from the dataframe to test my logic, and then I extended this to a function that could be applied to the dataframe as a whole with assistance from AI. From this, I saw that roughly 10% of calls were abandoned during the hold music out of the total abandoned calls that listened to hold music. Though not a focus of this analysis, I felt it was a chance to check that the values were reasonable and help understand if the scenarios I had identified were making up the majority of abandoned calls. Based on the suggestion in our mentor meeting, I checked if there were any calls that had an activity in their final row by filtering to only include the final row of the calls in the broader dataframe and then filtered the resulting dataframe for any values where there was an activity name. There turned out to be no rows which fit this criteria so this may help eliminate one possible abandonment scenario. From this, the next scenario I tried to identify was abandoned calls that never enter a queue, though at this point, I was not yet able to make the distinction between very brief calls that never enter a queue (e.g. LAC is closed so they should not be able to get in a queue; caller immediately hangs up) which appears to be neutral versus longer calls where the caller takes many steps but cannot find the proper menu and queue. To begin this, I found an example by finding the ‘Call ID’ value with the maximum number of instances as I thought this would reflect the call with the most steps in its journey, and I thought that many steps would indicate the user was unable to find the proper menu. Upon inspection, this example was an abandoned call where the caller seemed to rotate between Housing, Tenant, and PreTenant menus for roughly two hours. In terms of identifying abandoned calls that did not enter queues, I grouped the dataframe by ‘Call ID’ and applied a lambda function to check if the ‘Queue Name’ for all rows was NaN, then since I wanted to preserve the additional information in the dataframe, I took the ‘Call IDs’ from the grouped dataframe and used them to filter the original dataframe and found the number of calls by counting the unique Call IDs. There were 45,396 abandoned calls that never entered a queue which was significant compared to the 69,154 with the termination reason ‘Customer Left,’ but without further exploration, it cannot be said that these are all negative outcomes.

- How does it help the project?

This helps the project by developing a method to identify different categories of outcomes from the data, and works towards the goal of visualizing the different call outcomes by activity type so hopefully Legal Aid Chicago can better understand the factors that contribute to successful and unsuccessful call outcomes.

- Issues faced (if any)

There were not any major issues in this analysis, but I think as I continue working on this task, there will be the possibility for issues to arise if my set of scenarios for outcomes does not cover all possibilities, as well as the difficulty in possibly identifying where these gaps are within the data. Additionally, I began to run into the issue with identifying abandoned calls that never enter a queue because I was unsure of how to construct a criteria that could cover all possibilities as well as how granular my distinctions should get with regard to the different scenarios. For example, I did not see a clear metric for a differentiating between a short (neutral) call that does not enter a queue versus a long (negative) call that does not enter a queue, and I was not sure if a certain number of steps or duration of call would be a strong solution to this issue. I was also unsure about how detailed to go in identifying different scenarios as I know the goal is the three broad categorizations, but I wasn't sure if this should be balanced against including more detail about the activity for the visualization. Smaller issues included being unable to assign certain termination reasons to positive, negative, or neutral because I was unsure of their meaning ('RONA_TIMER_EXPIRED,' 'RONA Timer Expired,' 'CONTACT_CALLBACK_IN_PROGRESS'). However, they all seemed to have relatively few instances in the data so I felt okay in that they would not massively affect my analysis at this time (exact values are included at the very end of my code).

- Attempts to resolve issues (if any)

To resolve the difference between neutral and negative calls abandoned without entering a queue, I had initially tried to limit the duration of a call to a minute, assuming that I would adjust it depending on the trends I saw in the initial results. However, this had revealed that there were calls that were handled effectively (e.g. LAC closed and the user listens to the message then leaves) so I was unsure if this could miss out on insights based on the difference between the situations. For example, if a caller is so overwhelmed by hearing the initial message or attempting to begin navigating through the menus that they would rather hang up, I feel that is a negative outcome but I am unsure of if this can effectively be identified from the dataset without focusing too closely on any one example.

- Issues resolved (if any)

By adding Call ID and then sorting by it and Activity Start Timestamp, it resolved the issue of not being able to utilize consecutive dataframe rows to track the progression of a call's steps as previously successive steps of a call could be separated by any number of rows depending on the other calls occurring concurrently and their activity. This was useful in the development of a method to find calls that were abandoned while listening to the hold music. This further helped address issues of wanting to be able to group all entries of a call together without losing any information by using .groupby(). Additionally, I originally had issues with my function to find calls abandoned during the hold music as the time difference for the first row of each call was NaN, but this was easily resolved by filling these values with zero.

- Next steps

For next steps, I will refine the criteria used to identify the different categories of calls abandoned without entering a queue, and continue to develop methods for identifying the remaining abandonment scenarios from the data. I would also like to try and incorporate call outcome categorization or reason into the dataframe as a column as I feel it could help support visualization.

- References (mention if you built up on someone else's work)
    - Import file for CAR dataset
      https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Data%20import/CAR_data_import_code.ipynb
    - Meera's data cleaning
      https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Verbiage%20analysis/Meera%20Trivedi/Customer%20Abandonment_20Oct_Meera.ipynb

Presentation 1

- Links to presentation(s) and code(s) on GitHub

Presentation:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Data%20integrity%20checks/CARAllCallsIntegrity_7Oct_EDATeam2.pdf

Code:

https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Call%20outcome%20analysis/CallOutcomeAggregation_7Oct_Vivienne.ipynb

- What did you do?

My focus for this presentation was beginning to see if the CAR dataset could be utilized to replicate the Legal Menu Summary's information about call outcomes so that future analysis could rely only on the CAR dataset. I began by using the 'CAR_import_data' file to read in the data, and I also specifically created a folder that only contained the months where the CAR dataset included 'Termination Reason' as a column. I began by looking at the unique values for 'EP Name,' 'Queue Name' and 'Flow Name' to try to gauge what each step represented. My initial approach to tracking the call journey (in order to approximate which menu they had selected and which queue they entered; shown in Legal Summary Menu through the 'Legal Menu Option,' 'Menu Selection,' 'Queue Selection,' and 'Group suboption' columns) was to use the combinations of 'EP Name' and 'Queue Name' but I recognize this as a rough method until I can develop a more nuanced understanding of how the CAR data represents the call pathways. My first step was to create a dataframe with the total calls for each menu and queue combination by grouping by 'EP Name' and 'Queue Name' and then aggregating by unique 'Contact Session ID' values. I also filled the NaNs in the 'EP Name' and 'Queue Name' columns with 'Unknown' so that if there was a significant number of calls missing one of these columns, it could hopefully be used to help identify in which step or queue was causing issues. This was intended to begin recreating the ‘Total Calls’ column in Legal Menu Summary and also provide a baseline for comparison as I aggregated other call outcomes. For abandoned calls, I began by creating a subgrouping of the CAR dataset that only includes the periods where ‘Termination Reason’ was included as a column in the data as this creates the basis for determining the call outcome. I filtered the dataframe to only include the last instance of each ‘Contact Session ID’ as it seemed that this would be the best way to identify unique calls and the final step of the caller’s journey. However, I do recognize that there are potentially issues with this method depending on how call outcomes are being defined but I will discuss this further in the ‘Issues faced’ section. For open queue calls, I filtered the dataset for rows that had a ‘Queue Name’ value and did not have a ‘Termination Reason’ as I reasoned this would reflect ongoing calls where the caller was in a queue. The same method for aggregating total calls was applied to abandoned and open queue calls.

- How does it help the project?

This work helps the project because it helps further the goal of being able to conduct analysis on call outcomes while working to streamline the process by only utilizing one dataset. Additionally, it seems that the aggregation methods behind the Legal Menu Summary dataset are unclear so having an aggregation where the methods are known could help facilitate further analysis about trends in call outcomes.

- Issues faced (if any)

The main issue faced at this stage in analysis is understanding how these journeys through the different menus and queues are reflected in each dataset, and then how to effectively use that to replicate the summarized information from the Legal Menu Summary dataset with the CAR dataset. This also raised issues of how to define different outcomes based on the values available in the dataset, for example, ‘Customer Left’ could be taken as the customer abandoning their call, but it could also represent that they spoke to an agent and were successful with getting help with their legal issue. At this time, I felt I did not have enough familiarity with the dataset and the call flow to effectively define these outcomes through a combination of values. Additionally, there seems to be certain issues with the data itself and how it is being compiled or how the phone system itself is interpreting different actions. In the image below, it can be seen that the customer navigates through the menu and eventually selects receiving a courtesy callback about their issue. After several attempts to reach the customer, it seems that the agent and customer are both able to get on the call. However, after the customer hangs up, there are two more entries for the call (one at the same time as the customer is shown leaving, and the other two minutes later) but neither has an ‘EP Name’ or a ‘Termination Reason’ so if these were looked at as the last step in the call journey, all information about the call back and why the call ended would be lost. I recognize this issue is partially as result of my approach of only looking at the last instance of each ‘Contact Session ID,’ however, I’m unsure of how to systematically address this issue if it is a larger issue with nonconsecutive entries for a single call or if I just need a better understanding of the procedure used by the phone system to track and compile the steps in the journey.

492	3e0c8f9b-f617-4444-8b74-63d829f18c98	NaN	NaN	NaN	2025-03-17 08:03:32	Family	NaN	NaN	8
493	3e0c8f9b-f617-4444-8b74-63d829f18c98	All LAC Queues Telephony EP	NaN	NaN	2025-03-17 08:03:32	NaN	NaN	NaN	8
494	3e0c8f9b-f617-4444-8b74-63d829f18c98	All LAC Queues Telephony EP	NaN	NaN	2025-03-17 08:03:32	Family	NaN	NaN	8
495	3e0c8f9b-f617-4444-8b74-63d829f18c98	All LAC Queues Telephony EP	NaN	PreQueueMessage2	2025-03-17 08:03:32	NaN	NaN	NaN	8
496	3e0c8f9b-f617-4444-8b74-63d829f18c98	NaN	CourtesyCallback	NaN	2025-03-17 08:03:42	NaN	NaN	NaN	8
497	3e0c8f9b-f617-4444-8b74-63d829f18c98	Courtesy Callback Telephony EP	NaN	ReadANI	2025-03-17 08:03:42	NaN	NaN	NaN	8
506	3e0c8f9b-f617-4444-8b74-63d829f18c98	Courtesy Callback Telephony EP	NaN	CCB	2025-03-17 08:03:58	NaN	NaN	NaN	8
507	3e0c8f9b-f617-4444-8b74-63d829f18c98	NaN	NaN	NaN	2025-03-17 08:03:58	Family	NaN	NaN	8
508	3e0c8f9b-f617-4444-8b74-63d829f18c98	Courtesy Callback Telephony EP	NaN	PlayCCBConfirmation	2025-03-17 08:03:58	NaN	NaN	NaN	8
512	3e0c8f9b-f617-4444-8b74-63d829f18c98	Courtesy Callback Telephony EP	NaN	DisconnectContact1	2025-03-17 08:04:08	NaN	NaN	NaN	8
838	3e0c8f9b-f617-4444-8b74-63d829f18c98	Courtesy Callback Telephony EP	NaN	NaN	2025-03-17 08:14:20	Family	Marisol Guadarrama	Customer Left	8
840	3e0c8f9b-f617-4444-8b74-63d829f18c98	Courtesy Callback Telephony EP	NaN	LegalServerScreenPop	2025-03-17 08:14:21	NaN	NaN	NaN	8
849	3e0c8f9b-f617-4444-8b74-63d829f18c98	Courtesy Callback Telephony EP	NaN	NaN	2025-03-17 08:14:35	Family	Marisol Guadarrama	Customer Left	8
937	3e0c8f9b-f617-4444-8b74-63d829f18c98	Courtesy Callback Telephony EP	NaN	NaN	2025-03-17 08:21:06	Family	Marisol Guadarrama	Customer Left	8
938	3e0c8f9b-f617-4444-8b74-63d829f18c98	NaN	NaN	NaN	2025-03-17 08:21:06	Family	Marisol Guadarrama	NaN	8
970	3e0c8f9b-f617-4444-8b74-63d829f18c98	NaN	NaN	NaN	2025-03-17 08:23:06	Family	Marisol Guadarrama	NaN	8

I also apologize for modifying Objective 3 after our group's mentor meeting - it had not been my intention but as I was working, I just felt that it may be ineffective analysis as I was still unsure in my understanding of how to aggregate the CAR dataset so I felt it may be better approached in later work. I will be sure in the future to ensure that any potential changes to objectives are raised in the mentor meeting.

- Attempts to resolve issues (if any)

    At this point, I did not have any attempts to resolve the issues because I felt that my understanding of the dataset was not robust enough to effectively recognize and address the issues in the representation of the calls.

- Issues resolved (if any)

    I did not resolve any issues at this stage, but the confirmation that 'EP Name' refers to the entry point will help inform future work.

- Next steps

    For next steps, I would like to refine the approach used to define the menu-queue combinations to ensure that they are accurate to the pathways actually available to customers and the respective outcomes. I would also like to see if it is possible to create a systematic approach to identifying the last entry that is relevant to the outcome of interest, and then work to verify that the method is effective across all entries; this may involve further exploration of the data to identify if there are trends in how different types of entries appear or if it varies on a case-by-case basis.